

Trustworthy Machine Learning Assignment 3

Team number - LV

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Adversarial Robustness

For this assignment, we are supposed to train an image classification model that is robust to adversarial attacks. Our objective is to gain high performance on both clean and adversarially perturbed or attacked images. Combination of clean accuracy and robustness accuracy will be used as Evaluation Score, so the model maintains standard classification performance with opposition to adversarial perturbations.

The Process

Models required

We used the following neural network architectures-

- ResNet-18
- ResNet-34
- ResNet-50

We used the ResNet-50 architecture from torchvision.models and edited it by replacing the final fully connected layer so that it produces logits corresponding to the 9 classes.

Adversarial Training Methods

In order to improve robustness, following adversarial attack methods during training:

1) FGSM (Fast Gradient Sign Method)

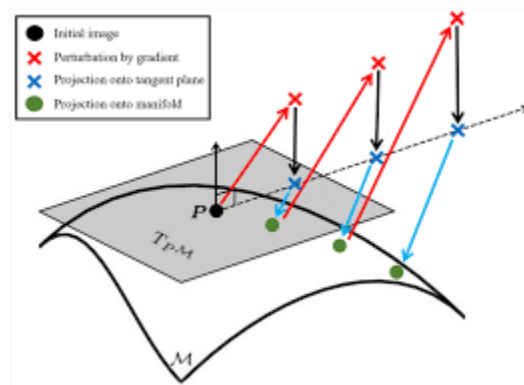
FGSM generates adversarial examples by adding a small perturbation in the direction of the gradient of the loss with respect to the input image. It basically takes direction and not magnitude to maximize Loss.

$$adv_x = x + \epsilon * \text{sign}(\nabla_x J(\theta, x, y))$$

This method is fast for adversarial training as it mixes and FGSM-perturbed images in each batch.

2) PGD (Projected Gradient Descent)

PGD is the strongest of white box attacks. It starts with a random perturbation and and executing multiple gradient ascent steps. After each step, the perturbation is clipped to remain within an ϵ -constrained region around the original image.



We used the three key hyperparameters in PGD as:

- $\epsilon = 8/255$ (perturbation budget)
- $\alpha = \epsilon \times 0.25 = 2/255$ (step size per PGD step)
- PGD steps = 14 (number of attack steps during training)

Validation robustness was evaluated using a stronger PGD attack with 20 steps to avoid overfitting to the exact training attack strength.

Model Trials

Before the final run, several settings were explored:

- Clean training: established an accuracy baseline ($> 80\%$) but zero robustness.
- FGSM training: faster than PGD but gave weaker robustness.
- ResNet-18 fine-tuning: starting from a trained checkpoint and fine-tuned with PGD. This improved robustness but showed an accuracy drop.

- ResNet-50 PGD from scratch (final): 90 epochs with the configuration above yielded the best combined score.
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Results

The final model used a ResNet-50 trained and trained on PGD adversarial for 90 epochs. It was then evaluated on a validation set comprising 10% of the training data. The model scored a clean accuracy of 0.7008 and a robust accuracy of 0.4818. Using the competitive scoring metric, the resulting Combined score was:

$$\begin{aligned}\text{Combined Score} &= (\text{Clean Accuracy} + \text{Robust Accuracy}) / 2 \\ &= 0.7008 + 0.4818 = 0.5913\end{aligned}$$

Comparing all evaluated models, the ResNet-50 trained with PGD adversarial training achieved the highest overall score of 0.5913. Models trained solely on clean data achieve higher accuracy on unperturbed inputs but remain highly at risk to adversarial attacks. On the contrary, adversarially trained models let go some clean accuracy in exchange for improved robustness.

The superior performance of the final ResNet-50 model also shows the results of stronger adversarial training procedures. Using multi-step PGD attacks during training gives more of a challenging optimization objective than single-step FGSM training, which then leads to significantly better opposition against adversarial perturbations.

Conclusion

We trained a ResNet-50 classifier using 90 epochs and trained it using PGD-based adversarial training and achieved a public leaderboard score of 0.5913. Neural networks are naturally at risk from adversarial attacks, where an attacker, either with white-box or only black-box access, can create inputs that seem identical to normal samples but can cause incorrect predictions. Such vulnerabilities are dangerous in real world, safety applications and softwares like autonomous driving, medical diagnosis, and security systems, where incorrect decisions can have big consequences.

Adversarial training is currently one of the most practical and effective defence mechanisms, but it has its drawbacks. Lack of Robustness checkup and accuracy along with high computational cost being some. This just shows that Robustness should be treated as one of the main objectives at the beginning of model creation rather than a later add-on. It also

highlights the importance of active research into Robustness methods to ensure safe and secure deployment of ML systems into real world environment domains.